Smart HR
Event-Driven Intelligent Workflow Automation Engine

Technical Project Report

Comprehensive System Documentation

Architecture | Design Patterns |

OOP Concepts | SOLID Principles

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React · Node.js · Express · PostgreSQL · Prisma · TypeScript

Event → Condition → Action Architecture

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1. Executive Summary

The Smart HR Event-Driven Intelligent Workflow Automation Engine is a premium, full-stack Human Resources management platform engineered to modernize and digitize traditionally fragmented HR workflows. By automating repetitive rule-based decisions — such as salary deductions, leave approvals, employee warnings, and bonus assignments — the platform delivers a cohesive, real-time, and transparent experience for all HR stakeholders.

1.1 Platform Overview

The platform serves three distinct user roles, each with tailored dashboards and workflows:

- **HR Admin / Manager** — Manages employee records, defines automation rules (Event → Condition → Action), configures multi-step workflows, triggers manual workflow runs, and views dashboard statistics.
- **Employee** — Marks daily attendance (which auto-triggers backend events), requests leaves, and views salary simulations reflecting applied rules.
- **Event Bus / Rule Engine** — The intelligent backend orchestrator that listens for domain events, fetches matching active rules, evaluates conditions using the Strategy Pattern, executes corresponding actions via the Factory and Command Patterns, and persists detailed action logs.

Mission Statement

The platform's mission is to eliminate manual, error-prone HR decision-making by delivering a secure, event-driven, and rule-configurable automation system that empowers HR teams to define business logic once and have it execute automatically and consistently across all employee lifecycle events.

1.2 Core Value Proposition

The system addresses several critical pain points in traditional HR management:

- **Event-Driven Automation** — Rules fire automatically when domain events occur (attendance marked, leave requested), eliminating manual processing delays.
- **Dynamic Rule Configuration** — HR Admins define conditions using flexible operators (greater than, less than, equals, contains) without requiring developer involvement.
- **Multi-Step Workflow Engine** — Complex sequential HR workflows (onboarding, appraisal cycles) are modelled as ordered steps with execution state tracking.
- **Role-Based Access Control** — Each stakeholder interacts only with features and data relevant to their organisational role.
- **Comprehensive Action Logging** — Every automated action is persisted with its trigger event, matched rule, result, and status for full auditability.

2. Technology Stack

The platform employs a modern, production-grade technology stack designed for scalability, type safety, and developer productivity. Every technology choice was deliberate, balancing performance, ecosystem maturity, and long-term maintainability.

Category	Technology	Purpose
Backend Runtime	Node.js + Express	High-performance async HTTP server with middleware pipeline
Language	TypeScript	End-to-end type safety across backend and frontend layers
ORM	Prisma ORM	Type-safe database client with automated migration support
Database	PostgreSQL	Relational database with full ACID transactional compliance
Frontend Framework	React 19	Component-based UI with concurrent rendering features
Build Tool	Vite	Ultra-fast HMR and optimised production bundling
Styling	Tailwind CSS v4	Utility-first styling with JIT compilation pipeline
Animations	Framer Motion	Declarative animation library for smooth UI transitions
State Management	Zustand	Lightweight global client state with minimal boilerplate
Server State	React Query	Caching, background sync, and async state management
API Communication	Axios	HTTP client with typed request/response interceptors
Validation	Zod	Runtime schema validation for all API requests and responses
Authentication	JWT + Cookies	Stateless auth with secure HttpOnly cookie storage
Icons	Lucide Icons	Consistent, open-source icon library for React

2.1 Backend Stack Deep Dive

The backend is built on Node.js with Express, leveraging TypeScript for complete type coverage across all layers. Prisma ORM abstracts raw SQL into a fully typed query interface, while PostgreSQL serves as the persistence layer with full ACID transactional support.

- **Express** provides a clean, middleware-first HTTP routing layer. Controllers remain thin by delegating all business logic to service classes, following strict SRP.
- **Prisma** generates a fully typed client from the schema definition, eliminating runtime type mismatches between the application and database layers entirely.
- **PostgreSQL** ensures data integrity through ACID compliance. The combination with Prisma enables type-safe, composable query building without raw SQL in application code.

2.2 Frontend Stack Deep Dive

The frontend is a React 19 single-page application built with Vite, combining Tailwind CSS for styling and Framer Motion for smooth UI transitions. Application state is split between Zustand for global client state and React Query for all server-synchronised data.

- **React Query** handles caching, background refetching, and stale-while-revalidate patterns, significantly reducing unnecessary API calls during navigation.
- **Zustand** provides a non-boilerplate-heavy alternative to Redux for sharing UI state across the component tree.
- **Framer Motion** enables declarative page transitions and micro-interactions that enhance the experience without sacrificing performance or bundle size.

3. System Architecture & Design

The platform follows a modular, layered architecture that cleanly separates concerns across feature domains. The system is designed around the principle of high cohesion within modules and low coupling between them, with the Event → Condition → Action pattern as the central architectural philosophy.

3.1 Backend Architecture

The backend is organised into feature modules, each encapsulating all related logic for a specific domain. This modular structure ensures that adding or modifying one feature domain does not impact unrelated modules.

Module	Responsibilities
Employee	Full CRUD management of employee records, department assignment, salary and status tracking
Attendance	Mark attendance, record check-in/check-out times, auto-publish attendance events to the EventBus
Leave	Leave request submission, approval lifecycle management, auto-trigger leave events
Rule Engine	Singleton orchestrator — listens to EventBus, fetches active rules, evaluates conditions, executes actions
Workflow	Multi-step workflow creation, step configuration, manual trigger execution, run state tracking
ActionLog	Persist every automated action with trigger event, matched rule, result JSON, and success/failure status
Dashboard	Aggregate statistics across employees, attendance, leaves, rules, and workflow runs for Admin view

3.1.1 Layered Architecture Pattern

Each backend module follows a strict three-layer pattern ensuring clean separation of HTTP, business, and data concerns:

- **Controller Layer** — Handles HTTP request parsing, response formatting, and route-level Zod validation. Controllers are intentionally thin and delegate all business logic to service classes immediately.
- **Service Layer** — Contains all business rules, orchestrates database operations, enforces domain constraints, manages cross-module interactions, and publishes events to the EventBus when domain state changes.
- **Data Layer (Prisma)** — Handles all database interactions through a type-safe Prisma client. Services interact only with Prisma models; no raw SQL exists anywhere in application code.

3.2 Event → Condition → Action Architecture

The system's defining architectural pattern is the Event → Condition → Action pipeline. When a domain event occurs (e.g., attendance is marked, a leave is requested), it flows through a structured processing pipeline:

- **Event** — A domain action persisted as an HREvent record (e.g., type: "attendance_marked") and published to the EventBus.
- **Condition** — The RuleEngine fetches all active rules matching the event type and passes each rule's condition (field, operator, value) to the Evaluator (Strategy Pattern) for boolean evaluation against the event payload.
- **Action** — If the condition evaluates to true, the ActionFactory (Factory Method Pattern) creates the appropriate Action Command (e.g., WarnEmployeeAction), which executes the configured business consequence.

3.3 Frontend Architecture

The frontend follows a component-based architecture with distinct organisational layers that separate UI rendering from data fetching and business logic:

- **Pages** — Top-level route components that compose smaller components and orchestrate all data fetching via React Query hooks connected to Axios API clients.
- **Components** — Reusable, stateless or locally stateful UI elements with clearly defined prop interfaces and no direct API dependencies.
- **Hooks** — Custom React hooks that encapsulate React Query calls, Zustand subscriptions, and derived state computations.
- **Services** — Axios-based API client modules that abstract all HTTP communication, providing typed request and response functions for each backend endpoint.

4. Database Design (ER Diagram)

The database schema is designed in PostgreSQL with Prisma, structured around seven core entities that reflect the real-world relationships within the HR workflow automation domain. The ER Diagram (Figure 1) illustrates all entities, their key fields, and the foreign-key relationships connecting them.

4.1 Key Schema Decisions

Entity	Key Fields	Relationships
Employee	id [PK], name, email, department, baseSalary, status	1-to-Many with Attendance, Leave, HREvent, ActionLog
Attendance	id [PK], employeeId [FK], date, checkIn, checkOut, status [Enum]	Many-to-One with Employee
Leave	id [PK], employeeId [FK], fromDate, toDate, status [Enum: PENDING/APPROVED]	Many-to-One with Employee
Rule	id [PK], name, eventType, conditionField, conditionOperator, conditionValue, actionType, actionConfig [JSON], isActive	Independent — matched by RuleEngine at runtime
HREvent	id [PK], type, employeeId [FK], payload [JSON], status	Many-to-One with Employee; 1-to-Many with ActionLog
ActionLog	id [PK], eventId [FK], ruleId [FK], employeeId [FK], actionType, result [JSON], status	Many-to-One with HREvent, Rule, Employee
Workflow	id [PK], name, description, isActive	1-to-Many with Step; 1-to-Many with WorkflowRun
Step	id [PK], workflowId [FK], name, type, config [JSON], order	Many-to-One with Workflow; 1-to-Many with StepRun
WorkflowRun	id [PK], workflowId [FK], status, startedAt, completedAt	Many-to-One with Workflow; 1-to-Many with StepRun
StepRun	id [PK], workflowRunId [FK], stepId [FK], status, result [JSON]	Many-to-One with WorkflowRun and Step

Figure 1: Entity-Relationship (ER) Diagram — Smart HR Workflow Automation Engine Database Schema (see attached diagram in Appendix A)

4.2 Enum Definitions

Enum	Values	Used In
AttendanceStatus	PRESENT, ABSENT, LATE, HALF_DAY	Attendance.status
LeaveStatus	PENDING, APPROVED, REJECTED	Leave.status
EmployeeStatus	ACTIVE, INACTIVE	Employee.status
HREventStatus	PENDING, PROCESSED, FAILED	HREvent.status
ActionLogStatus	SUCCESS, FAILED, SKIPPED	ActionLog.status
WorkflowRunStatus	RUNNING, COMPLETED, FAILED	WorkflowRun.status

5. System Use Cases & Actor Interactions

The use case diagram (Figure 2) illustrates the complete interaction surface of the platform across its three primary actors: HR Admin/Manager, Employee, and the Event Bus / Rule Engine. Extended use cases (shown with dashed arrows) represent conditional or automated behaviours.

Figure 2: Use Case Diagram — Smart HR Workflow Automation Engine Actor Interactions (see attached diagram in Appendix A)

5.1 HR Admin / Manager Actor

- **Manage Employees (CRUD)** — Create, read, update, and delete employee records including department and salary details.
- **Define Automation Rules** — Configure Event, Condition, and Action triplets to encode business logic as configurable rules without code changes.
- **Manage Multi-step Workflows** — Design sequential workflow templates with ordered steps and action configurations for complex HR processes.
- **Trigger Manual Workflow Runs** — Initiate on-demand workflow executions outside the automatic event-driven pipeline.
- **View Dashboard Statistics** — Access aggregate metrics on employees, attendance patterns, leave trends, and rule execution history.

5.2 Employee Actor

- **Mark Attendance** — Records check-in/check-out; automatically publishes an "attendance_marked" event to the EventBus, triggering rule evaluation.
- **Request Leave** — Submits a leave application; auto-triggers a "leave_requested" event for rule-based auto-approval or escalation.
- **View Salary Simulation** — Reviews personalised salary calculations reflecting any deductions or bonuses applied by automated rules.

5.3 Event Bus / Rule Engine Actor

- **Listen to Event Bus** — Subscribes to all published domain events using the Observer Pattern.
- **Fetch Relevant Rules** — Queries the RuleRepository for all active rules whose eventType matches the incoming event type.
- **Evaluate Condition Logic** — Delegates each rule's condition to the Evaluator (Strategy Pattern) which resolves the appropriate operator at runtime.
- **Execute Action Commands** — If a condition is satisfied, the ActionFactory creates and executes the appropriate Action Command.
- **Persist Action Logs** — Writes a complete ActionLog record for every processed rule, capturing the trigger event, matched rule, execution result, and final status.

6. Object-Oriented Programming Concepts

The platform is architected with a deep application of OOP principles, ensuring that real-world HR domain concepts (employees, rules, events, workflows, actions) are modelled as coherent, reusable, and maintainable software entities.

6.1 Encapsulation

Business logic and data are encapsulated within well-defined service classes, preventing unauthorised direct access and ensuring that changes to implementation details do not leak across module boundaries.

- **RuleEngine** — Encapsulates the complete event processing pipeline: subscription, rule fetching, condition evaluation, action execution, and log persistence. External code simply publishes events without knowing the internal orchestration sequence.
- **ActionFactory** — Encapsulates the action instantiation registry. Consumers call create(actionType) and receive a ready-to-execute action without knowledge of which concrete class is returned.
- **Evaluator** — Encapsulates all condition evaluation logic and operator strategy selection. The RuleEngine calls evaluate(condition, payload) and receives a boolean result without knowing which operator was applied.
- **WorkflowService** — Encapsulates the full workflow execution lifecycle from step sequencing to run state management and result persistence.

6.2 Abstraction

Complexity is hidden behind clean, simple interfaces. High-level components interact with abstractions rather than concrete implementations, enabling internal complexity to grow without affecting consumers.

The RuleEngine calls evaluator.evaluate(condition, eventPayload) and receives a boolean. It has no awareness of whether the evaluation uses a strategy map, switch-case, or any other mechanism internally. This abstraction boundary means the evaluation logic can be completely replaced without touching the engine's orchestration code.

6.3 Inheritance & Polymorphism

The Action Command hierarchy uses polymorphism extensively. All concrete action classes extend a common BaseAction abstract class, allowing the RuleEngine to call execute(event, rule) uniformly regardless of which specific action type is being executed at runtime.

- **DeductSalaryAction**, **WarnEmployeeAction**, **AutoApproveLeaveAction**, **AddBonusAction**, and **EscalateAction** all extend BaseAction and are substitutable interchangeably through the abstract interface.
- The Evaluator's strategy map values are all functions with an identical (a, b) → boolean signature, enabling polymorphic dispatch through a string-keyed registry.

6.4 Composition Over Inheritance

The system preferentially uses composition to build complex behaviours from simple, reusable parts rather than deep inheritance hierarchies that create tight coupling.

- **RuleEngine** composes EventBus, RuleRepository, Evaluator, and ActionFactory objects rather than inheriting from any base orchestrator class.
- **Evaluator** composes a registry (Map) of operator functions rather than inheriting from operator base classes, enabling dynamic registration of new operators at runtime.

7. SOLID Design Principles

The SOLID principles are systematically applied throughout the codebase, ensuring the system remains maintainable, extensible, and testable as it grows in complexity.

Letter	Principle	Core Idea
S	Single Responsibility	Every class has exactly one reason to change
O	Open/Closed	Open for extension, closed for modification
L	Liskov Substitution	Subtypes are fully substitutable for their base types
I	Interface Segregation	Clients depend only on the methods they actually use
D	Dependency Inversion	Depend on abstractions, never on concrete implementations

7.1 S — Single Responsibility Principle (SRP)

Every class has a single, well-defined responsibility. No class is responsible for more than one concern:

- **Controllers** — Exclusively handle HTTP request/response lifecycle. No business logic lives inside controllers.
- **Services** — Exclusively handle business rules and data orchestration. No HTTP-specific code exists inside services.
- **RuleEngine** — Exclusively orchestrates the event processing pipeline. It does not handle HTTP, database schema migrations, or UI concerns.
- **Concrete Action Classes** — Each action class (WarnEmployeeAction, DeductSalaryAction) is responsible only for executing its specific action type, nothing else.

7.2 O — Open/Closed Principle (OCP)

The system is open for extension but closed for modification. New capabilities can be added without altering existing, tested code.

Action Extension Example:

Adding a new "SendPayslip" action requires only creating a new SendPayslipAction class extending BaseAction and registering it in the ActionFactory registry. The RuleEngine, all existing actions, the Evaluator, and all existing tests remain completely untouched.

- New automation rules are new Rule records in the database — no code changes required.
- New condition operators are new entries in the Evaluator's strategy registry — no switch/case blocks to modify.
- New workflow step types are new handler functions — the workflow execution engine remains unchanged.

7.3 L — Liskov Substitution Principle (LSP)

All concrete action implementations can be used interchangeably through the BaseAction abstract class. The RuleEngine never checks which concrete action class it holds — it simply calls execute(event, rule) and trusts the correct behaviour is delivered.

- **WarnEmployeeAction.execute()** sends a warning notification to the employee.
- **DeductSalaryAction.execute()** applies a salary deduction to the employee record.
- **AutoApproveLeaveAction.execute()** marks the leave request as approved automatically.

All three conform to the BaseAction.execute(event, rule) contract — the substitution is seamless and the RuleEngine remains oblivious to the implementation details.

7.4 I — Interface Segregation Principle (ISP)

The system uses specific, narrow interfaces and Data Transfer Objects (DTOs) so that consuming classes depend only on the subset of the interface they actually need.

- **CreateEmployeeDTO** contains only the fields required for employee creation — not the full Employee model with auto-generated fields.
- **CreateRuleDTO** contains only the fields required to define a new automation rule — not audit or runtime fields.
- Service interfaces expose only the methods relevant to their consumers, avoiding god-object interfaces that force clients to depend on unused operations.

7.5 D — Dependency Inversion Principle (DIP)

High-level modules (Controllers) depend on abstractions (Service interfaces), not on concrete database implementations. This decoupling enables service implementations to be swapped or mocked without affecting the consuming controller.

- **EmployeeController** depends on IEmployeeService, not on EmployeeService directly.
- **RuleEngine** depends on IRuleRepository, not on the concrete Prisma implementation, enabling the repository to be swapped for a Redis-backed or in-memory variant in tests.
- This abstraction chain means unit tests can inject mock services into controllers without spinning up a PostgreSQL database at all.

8. Design Patterns

The platform implements six well-established Gang of Four design patterns, each chosen to solve a specific architectural challenge in a clean, idiomatic TypeScript way.

8.1 Observer Pattern — EventBus (Real-Time Event Distribution)

Location: backend/src/core/EventBus.ts

The EventBus implements the Observer Pattern to decouple event producers (Controllers/Services) from event consumers (RuleEngine). When a domain event occurs, the producing service publishes it to the EventBus without knowing which consumers are subscribed.

- **subscribe(topic, listener)** — Registers a listener function for a specific event topic.
- **publish(event)** — Notifies all registered listeners for the event's type, passing the full event payload.
- **unsubscribe(topic, listener)** — Removes a previously registered listener, preventing memory leaks.

Why Observer Pattern? Without the Observer Pattern, every service that creates a domain event would need to explicitly call the RuleEngine, creating tight coupling. With EventBus, producers and consumers are completely decoupled — new consumers can be added without any change to producers.

8.2 Singleton Pattern — RuleEngine, EventBus & PrismaClient

Location: backend/src/core/RuleEngine.ts, backend/src/config/database.ts

The RuleEngine, EventBus, and PrismaClient are each instantiated once as module-level singletons. All service classes import and share these single instances, ensuring consistent state and preventing resource duplication.

- **RuleEngine Singleton** — Ensures exactly one orchestrator processes all events, preventing duplicate rule evaluations for the same event.
- **EventBus Singleton** — Ensures all producers publish to the same event channel and all consumers receive every event exactly once.
- **PrismaClient Singleton** — Prisma maintains an internal connection pool. Creating multiple PrismaClient instances would open multiple pools, exhausting PostgreSQL connections. The Singleton guarantees exactly one pool for the application's lifetime.

8.3 Strategy Pattern — Condition Evaluator

Location: backend/src/core/Evaluator.ts

The Evaluator uses the Strategy Pattern to select and apply the correct condition evaluation algorithm at runtime based on the conditionOperator field of each Rule. Operators are stored in a Map registry and resolved dynamically.

Operator	Strategy Function	Example Condition
gt	a > b	hoursWorked > 8

lt	a < b	hoursWorked < 4
eq	a === b	status === "ABSENT"
neq	a !== b	department !== "Engineering"
contains	a.includes(b)	leaveReason contains "medical"
gte	a >= b	attendanceCount >= 20
lte	a <= b	baseSalary <= 30000
in	arr.includes(a)	department in ["HR","Finance"]

8.4 Factory Method Pattern — ActionFactory

Location: backend/src/core/ActionFactory.ts

The ActionFactory implements the Factory Method Pattern to create action command instances based on the actionType string from a matched Rule. The factory maintains a registry mapping type strings to concrete Action class constructors.

actionType	Created Action	Behaviour
warn_employee	WarnEmployeeAction	Logs a formal warning against the employee record
deduct_salary	DeductSalaryAction	Applies a configured percentage deduction to base salary
auto_approve_leave	AutoApproveLeaveAction	Automatically approves the leave request without manual review
add_bonus	AddBonusAction	Credits a bonus amount to the employee's salary record
send_notification	SendNotifAction	Sends an automated notification to the employee
escalate	EscalateAction	Escalates the event to the HR Manager for manual review
generate_coupon	GenerateCouponAction	Generates a reward coupon for high-performing employees

8.5 Command Pattern — BaseAction & Concrete Actions

Location: backend/src/core/actions/

The Command Pattern is implemented through the BaseAction abstract class, which defines a uniform execute(event, rule) interface for all action types. This separates the action request (the Rule match) from the action execution (the concrete class behaviour).

• **BaseAction** — Abstract class with abstract execute(event, rule): Promise method and a protected log(result) utility for writing ActionLog records.

• All concrete actions extend BaseAction and implement execute() with their specific domain behaviour, while inheriting the log() utility automatically.

• The RuleEngine invokes action.execute(event, rule) uniformly for any action type, following the Command Pattern's principle of decoupled invocation.

8.6 Repository Pattern — RuleRepository

Location: backend/src/repositories/RuleRepository.ts

The RuleRepository implements the Repository Pattern to abstract all database access operations for Rule entities away from the RuleEngine orchestrator. This separation ensures the engine's logic is completely independent of the persistence mechanism.

• **findActiveRulesByEvent(eventType)** — Returns all active Rule records matching the given event type in a single optimised query.

• **findById(id)** — Retrieves a single rule by primary key for management operations.

• The repository can be swapped for a Redis-cached or in-memory implementation without any change to the RuleEngine's code.

9. Key Features & Modules

9.1 Intelligent Rule Engine

The Rule Engine is the core of the platform, implementing a sophisticated event-driven automation pipeline that processes domain events against configurable business rules in real time.

- **Event Subscription** — Subscribes to all domain event types on initialisation, ensuring no event is missed regardless of source.
- **Active Rule Fetching** — Queries the RuleRepository for all enabled rules matching the incoming event type, supporting multiple rules per event.
- **Condition Evaluation** — Delegates condition evaluation to the Evaluator (Strategy Pattern), supporting eight configurable operators without any switch-case logic in the engine.
- **Action Execution** — Uses ActionFactory to create and execute the appropriate action command when a condition is satisfied.
- **Action Log Persistence** — Writes a detailed ActionLog record for every rule processed (success, failure, or condition-not-met) for full auditability.

9.2 Multi-Step Workflow Engine

The Workflow module provides a structured system for modelling complex, sequential HR processes that go beyond single-event rule evaluations.

- **Workflow Templates** — HR Admins define reusable Workflow entities containing an ordered sequence of Steps with individual type and configuration settings.
- **WorkflowRun Tracking** — Each manual or automated trigger creates a WorkflowRun record, with StepRun records tracking the execution state and result of every individual step.
- **Manual Trigger Endpoint** — HR Admins can trigger workflow execution on demand via a dedicated API endpoint, independent of the automatic event pipeline.

9.3 Employee & Attendance Management

- **Employee CRUD** — Full lifecycle management including creation, profile updates, department reassignment, salary adjustments, and status changes.
- **Attendance Marking** — Employees mark attendance with optional check-in/check-out timestamps; the service automatically publishes an "attendance_marked" HREvent to trigger rule evaluation.
- **Leave Management** — Employees submit leave requests; the service publishes a "leave_requested" HREvent, enabling automatic approval or escalation based on configured rules.

9.4 Role-Based Access Control (RBAC)

The platform enforces strict role separation through JWT-decoded role claims verified by server-side middleware on every protected route.

- **Employees** — Access limited to marking attendance, submitting leave requests, and viewing their own salary simulations.

- **HR Admin / Manager** — Full access to employee management, rule configuration, workflow management, dashboard analytics, and action log review.
- **authGuard Middleware** — Verifies JWT signature and expiry on every protected route before allowing the request to reach the controller.
- **roleGuard Middleware** — Verifies that the decoded JWT role claim matches the required role for the specific endpoint, returning 403 Forbidden on violation.

10. Sequence Diagram — Core Workflow

The sequence diagram (Figure 3) illustrates the complete end-to-end flow of the system's core use case: an Employee marking attendance, which triggers an automated rule evaluation and action execution through the full Event → Condition → Action pipeline.

Step	ParticipantFlow	Description
1	Employee → API	POST /api/attendance — Employee submits attendance mark request
2	Controller → Prisma DB	prisma.attendance.create(data) — Attendance record saved to PostgreSQL
3	Controller → Prisma DB	prisma.hREvent.create({ type: "attendance_marked" }) — HREvent persisted
4	Controller → EventBus	eventBus.publish("attendance_marked", event) — Event dispatched to all subscribers
5	EventBus → RuleEngine	RuleEngine.processEvent(event) — Observer callback triggered (Singleton)
6	RuleEngine → RuleRepo	ruleRepo.findActiveRulesByEvent("attendance_marked") — Active rules fetched
7	RuleEngine → Evaluator	evaluator.evaluate(condition, eventPayload) — Strategy Pattern resolves operator (e.g. hoursWorked < 8)
8	RuleEngine → ActionFactory	ActionFactory.create("warn_employee") — Factory Method creates WarnEmployeeAction
9	ActionFactory → DB/Notify	action.execute(event, rule) — Command executes: updates DB or sends notification
10	RuleEngine → Prisma DB	prisma.actionLog.create({ status, result }) — Action outcome persisted as ActionLog
11	API → Employee	200 OK "Attendance Marked" — Response returned to Employee

Figure 3: Sequence Diagram — Event-Driven Rule Execution Flow (see attached diagram in Appendix A)

Design Patterns Active in This Flow

Step	Pattern Applied	Component
4	Observer Pattern	EventBus.publish() notifies all subscribers without tight coupling
5	Singleton Pattern	RuleEngine — single instance processes all events globally
6	Repository Pattern	RuleRepository.findActiveRulesByEvent() abstracts DB queries

7	Strategy Pattern	Evaluator selects and applies the correct operator at runtime
8	Factory Method Pattern	ActionFactory.create(actionType) returns the correct Action object
9	Command Pattern	BaseAction.execute(event, rule) provides uniform invocation interface

11. Class Diagram — Architecture

The class diagram (Figure 4) models the complete backend architecture across four distinct layers: REST API (Controllers), Core Workflow Engine, Evaluation & Execution, and Database Access. The diagram highlights the relationships and design pattern implementations across all classes.

Figure 4: Class Diagram — Backend Architecture (see attached diagram in Appendix A)

11.1 Class Hierarchy Summary

Layer	Classes	Pattern(s)
REST API Layer	AttendanceController, EmployeeController, RuleController, WorkflowController, DashboardController	SRP — thin controllers, delegates to services
Core Workflow Engine	EventBus, RuleEngine, RuleRepository, Evaluator, ActionFactory	Observer, Singleton, Repository, Strategy, Factory
Evaluation & Execution	BaseAction (abstract), DeductSalaryAction, WarnEmployeeAction, AutoApproveLeaveAction, AddBonusAction, SendNotifAction, EscalateAction	Command Pattern — uniform execute() interface
Database Access Layer	PrismaClient (Singleton), Employee, Attendance, Leave, Rule, HREvent, ActionLog, Workflow, Step, WorkflowRun, StepRun	Singleton, Repository, Prisma Delegates

11.2 Key Relationships

- **RuleEngine** → **EventBus** — Subscribes to all domain events (Observer).
- **RuleEngine** → **RuleRepository** — Fetches active rules by event type (Repository).
- **RuleEngine** → **Evaluator** — Delegates condition evaluation (Strategy).
- **RuleEngine** → **ActionFactory** — Creates action instances on condition match (Factory).
- **ActionFactory** → **BaseAction** — Produces Command objects from the registry.
- **Concrete Actions** → **BaseAction** — Inheritance relationship: all actions extend BaseAction (Command).
- **RuleRepository** → **PrismaClient** — Uses the Singleton Prisma client for all DB queries.
- **Controllers** → **Services** → **PrismaClient** — Full three-layer dependency chain with DIP applied throughout.

12. Security & Performance

12.1 Security Architecture

- **JWT Authentication** — Tokens are signed with a server-side secret and stored in HttpOnly cookies, preventing XSS attacks from accessing authentication credentials through JavaScript.
- **Zod Schema Validation** — All incoming request bodies are validated against strict Zod schemas at the middleware layer before reaching controllers, preventing malformed or malicious data from entering business logic.
- **Role Guards** — Middleware verifies JWT role claims on every protected route, ensuring Employees cannot access Admin endpoints and vice versa.
- **Password Hashing** — Employee passwords are hashed with bcrypt before storage. Plaintext passwords never persist in the database at any point.
- **Input Sanitisation** — All actionConfig JSON fields are parsed and validated before execution to prevent injection of arbitrary code through rule configurations.

12.2 Performance Optimisations

- **React Query Caching** — Employee lists, rule configurations, and dashboard data are cached client-side with configurable stale times, reducing redundant API calls during normal navigation.
- **Prisma Query Optimisation** — Only required fields are selected in database queries using Prisma's select API, avoiding over-fetching of large JSON payload fields like actionConfig and result.
- **Singleton Database Client** — A single Prisma connection pool serves all service requests, maximising PostgreSQL connection efficiency and preventing pool exhaustion.
- **EventBus Topic Scoping** — Events are published with specific topic strings, ensuring listeners are only invoked for relevant event types rather than broadcasting all events to all listeners.
- **Vite Build Optimisation** — Tree-shaking, code splitting, and asset optimisation are applied at build time, minimising JavaScript bundle sizes for fast initial page loads.

13. System Design Principles

Beyond SOLID and OOP, the platform adheres to broader system design principles that ensure long-term reliability, observability, and maintainability.

13.1 Separation of Concerns (SoC)

Every layer of the system has a distinct, non-overlapping responsibility. The HTTP layer knows nothing about rule evaluation logic. The RuleEngine knows nothing about HTTP request formats. The database layer knows nothing about business rules or event routing.

13.2 DRY — Don't Repeat Yourself

Common logic is extracted into shared utilities and middleware. JWT verification, Zod validation, error formatting, and ActionLog persistence are each defined once and reused consistently across all modules without duplication.

13.3 Fail Fast & Validate Early

Zod validation middleware rejects malformed requests at the network boundary, before any business logic executes. This prevents invalid data from propagating through the system and ensures error messages are generated at the earliest possible point in the pipeline.

13.4 Event-Driven Architecture

The core automation pipeline is fully event-driven via the EventBus. State changes (attendance marked, leave requested) trigger events that propagate to all interested consumers without tight coupling — producers do not know which consumers are listening, enabling consumers to be added or removed without any producer code change.

13.5 Stateless API Design

The REST API is stateless — each request contains all information needed to process it (via JWT in cookie). No server-side session state is maintained, enabling horizontal scaling of the backend without session affinity requirements.

14. Development Status & Future Roadmap

14.1 Current Progress

The platform has completed its foundational milestone with the following capabilities fully operational:

- PostgreSQL database migration complete with full Prisma schema across all ten entity models.
- EventBus Observer pattern fully implemented and integrated with all domain event publishers.
- RuleEngine Singleton with Condition Evaluator (Strategy Pattern) and ActionFactory (Factory Method) operational.
- All six concrete Action Command classes implemented and registered in the ActionFactory registry.
- JWT authentication with authGuard and roleGuard middleware fully active on all protected routes.
- Multi-step Workflow Engine with WorkflowRun and StepRun execution tracking deployed.
- Employee, Attendance, and Leave modules with automatic HREvent publishing on state changes.
- Dashboard statistics aggregation endpoint live for HR Admin overview.
- Complete ActionLog persistence for every automated rule execution.

14.2 Future Roadmap

Feature	Description
Real-Time Notifications	WebSocket or Server-Sent Events integration to push rule execution results and workflow status updates to the HR Admin dashboard in real time.
AI-Assisted Rule Suggestions	ML-based recommendation engine analysing historical ActionLog data to suggest new automation rules based on HR patterns and anomalies.
Mobile Application	React Native port of the Employee-facing interface for iOS and Android, enabling mobile attendance marking and leave requests.
Advanced Analytics Dashboard	HR revenue dashboards with attendance heatmaps, rule trigger frequency analysis, department-wise automation insights, and exportable reports.
Workflow Visual Designer	Drag-and-drop visual workflow builder allowing HR Admins to design multi-step workflows without JSON configuration.
Multi-Tenancy Support	Organisation-level data isolation to support multiple companies on a single platform deployment with complete data segregation.
Audit Trail & Compliance	Immutable audit log for all HR actions, rule changes, and data modifications to support regulatory compliance requirements.
Integration Webhooks	Outbound webhook system allowing ActionLog events to trigger external HRMS, payroll, or notification systems via configurable HTTP endpoints.

15. Conclusion

The Smart HR Event-Driven Intelligent Workflow Automation Engine represents a well-architected, full-stack solution to the real-world problem of manual, rule-based HR decision-making. Its technical depth is evidenced by the deliberate application of OOP principles, SOLID design guidelines, and six proven Gang of Four design patterns across a modern TypeScript codebase.

The Observer Pattern via the EventBus brings clean, decoupled event distribution across all domain modules. The Singleton Pattern ensures resource-efficient access to the RuleEngine, EventBus, and database client. The Strategy Pattern in the Evaluator enables flexible, runtime-configurable condition evaluation without any switch-case proliferation. The Factory Method Pattern in the ActionFactory allows new action types to be added without modifying any orchestration code. The Command Pattern through BaseAction provides a uniform execution interface across all action implementations. The Repository Pattern abstracts all database queries from the engine's orchestration logic.

Together, these patterns form a codebase that is not only functionally complete today but designed to scale, extend, and evolve with minimal technical debt. The system serves equally well as a portfolio showcase, a learning reference for advanced design patterns in TypeScript, and a solid foundation for a production-grade commercial HR automation product.

Technical Excellence Summary

The Smart HR Workflow Automation Engine achieves a rare combination of pedagogical clarity and production readiness — demonstrating OOP, SOLID, and system design principles in concrete, working implementations rather than abstract theory. Six Gang of Four patterns are applied systematically across a TypeScript full-stack architecture anchored by the Event → Condition → Action pipeline, delivering both an exemplary academic submission and a commercially viable HR automation foundation.

Appendix A: System Diagrams

The following pages contain all four UML architectural diagrams generated from the codebase analysis. Each diagram is produced at high resolution for print and digital use.

Reference	Diagram	Description
A.1	**Use-Case Diagram**	Complete actor interactions across HR Admin, Employee, and Event Bus / Rule Engine actors with all use cases and extend/include relationships.
A.2	**ER Diagram**	Full entity-relationship diagram showing all ten Prisma models, their fields (with PK/FK annotations), enum types, and all foreign-key relationships.
A.3	**Sequence Diagram**	End-to-end sequence flow for the core "Mark Attendance → Rule Evaluation → Action Execution" pipeline across all eight participating components.
A.4	**Class Diagram**	Complete class hierarchy across all four architectural layers, showing attributes, methods, stereotypes, design pattern badges, and all inter-class relationships.

Note: The four PNG diagram files (UseCase_Diagram.png, ER_Diagram.png, Sequence_Diagram.png, Class_Diagram.png) are provided as separate high-resolution files alongside this report and should be appended or inserted into the final submission document.